

## **SYLLABUS**

### **SYSTEM ENGINEERING IN PUBLIC PROGRAMS**

#### **PRISCILA DE AZEVEDO DRUMMOND**

Welcome to Systems Engineering in Public Programs! This course invites you to explore how mathematical modeling can serve the public good. Together, we'll tackle real-world challenges that affect communities, examining how different stakeholders shape solutions and how our modeling choices impact society. Your diverse perspectives and experiences will enrich our collective understanding of these complex systems.

#### **2025 FALL**

**Start and End Dates:** September 06th, 2025 to December 14th , 2025

**Credit Hours:**

**Location:** EBT Canvas Course - Priscila De Azevedo Drummond

**Course Format:** Face-to-face (15-week semester)

#### **Instructor Information**

Priscila de Azevedo Drummond (she/her)

deazevedodrummond.p@northeastern.edu

Office Hours/Virtual office hours: Every Monday from 6:00 pm to 7:00 pm and every Thursday from 11 am to 12 am via Zoom – link (Meeting ID withheld). If you want to come to the meeting just to have an accountability partner for your work, you are also welcome.

You are welcome to contact me at any time via email or Teams. I usually check them only once a day, from Monday to Saturday, by 10 am. Please, always mention the class you are in, and the project group you are part of to facilitate communication.

If I don't get to respond in 48 hours (from Monday to Saturday), feel free to send me a reminder.

#### **Diversity Statement**

You belong here. Each of you brings valuable perspectives shaped by your unique experiences, identities, and backgrounds, including but not limited to race, ethnicity, gender identity, sexual orientation, ability status, age, socioeconomic background, nationality, and religious or cultural traditions. These diverse viewpoints are not just welcomed; they are essential to our collective learning, especially in a field dedicated to serving diverse public communities.

In this course, we commit to creating a safe space where we can engage with challenging ideas while treating each other with respect and care. Systems engineering for public programs requires us to understand multiple stakeholder perspectives. This starts in our classroom. We will encounter complex topics that may evoke strong emotions or differing opinions. In these moments, we practice **active listening**, assume **positive intent**, and **challenge ideas rather than individuals**.

If you encounter any aspect of the course, whether content, teaching methods, or classroom dynamics, that create barriers to your full participation or learning. I invite you to communicate with me. This can be

through office hours, email, or anonymous feedback. Your success matters, and I am committed to working with you to make adjustments that support your learning journey.

Together, we create a community where everyone can engage authentically, take intellectual risks, and grow as engineers committed to equitable solutions for society's complex challenges.

### Course Prerequisites

Python basic knowledge – we are using the Claude.ai to upgrade our coding capabilities during the course. If you don't know the language yet, contact me before the course starts.

Engineering Statistics or equivalent

Operations Research or equivalent

### Course Description

This course focuses on the definition, development, and mock-up implementation of mathematical models in public interest contexts. You'll learn to navigate multiple stakeholder perspectives and collaborative decision-making in complex public settings. Through realistic case studies, you'll apply modeling techniques including Game Theory, Cost-Benefit Analysis, Bayesian Decision Process, DEA, Simulation, Queuing Theory, Multi-criteria Decision-making, and Differential Equations to address societal challenges.

### Course Learning Goals and Objectives

By the end of this course, you will be able to:

**Design** mathematical problems using appropriate assumptions and strategies based on different stakeholders' needs

**Evaluate** whether model solutions prioritize feasibility or optimality

**Select** modeling tools based on problem requirements and justify your tool selection

**Adapt** decision-model strategies to different contexts

**Critically discern** how modeling choices might have unintended consequences

*These aren't just academic skills – they are capabilities that will serve you throughout your career in addressing how to apply OR/IE to complex societal challenges.*

### Required Course Materials

There is no required textbook for this course. Instead, you will be expected to read select book chapters and journal articles throughout the semester. Most will be posted to Canvas

All the material used will be uploaded from open source or journal papers. Some will be based on problems from resources available online through the Northeastern Library including

- Community-Based Operations Research (2012)
- Operations Research and Healthcare Policy (2013)
- Handbook of Operations Research for Homeland Security (2013)

- Operations Research, Taha (2007)
- Socio-Economic Planning Journal

### Required Course Hardware and Support

- Reliable internet connection (please, make use of the university facilities whenever you need, and let me know if you need any additional support)
- Access to computer with Python/Jupyter Notebook capability

Prior to taking this course, review the basic computer specifications for Canvas (Links to an external site.). You can access Canvas on the go using the Canvas Student app (Links to an external site.). Not all features work on the mobile app, I recommend completing essential course work using your computer and a web browser.

If you need help to using Canvas, online chat and phone support are available 24/7 by clicking on the help menu in the left-hand navigation. For additional help, around-the-clock Northeastern tech support is still available students. For technology questions or assistance, contact the IT Service Desk 24 hours a day, 7 days a week, learn about ways to contact ITS Service Desk here (Links to an external site.).

### Course Methodology

This course embraces project-based learning and active learning through case studies, collaborative problem-solving, and iterative project work. You'll engage with material through:

**Collaborative discussions** where diverse viewpoints strengthen solutions

**Hands-on modeling** using Jupyter notebooks and real data

**Stakeholder simulations** to understand multiple perspectives

**Computer simulations** to understand difference of feasible and optimal spaces in practice

**Peer learning** through tutorials and group work

**Reflective practice** to examine the implications of your choices

*Mistakes are learning opportunities and are expected to happen. Questions are welcomed and encouraged.*

### Grading/Evaluation Standards

| 94 – 100 | A  | 77 - 89 | B+ | 77 – 79 | C+ | 67 – 69 | D+ |        |   |
|----------|----|---------|----|---------|----|---------|----|--------|---|
| 90 – 93  | A- | 84 – 86 | B  | 74 – 76 | C  | 64 – 66 | D  | 0 – 59 | F |
|          |    | 80 – 83 | B- | 70 – 73 | C+ | 60 – 63 | D- |        |   |

[If you use a grading rubric, you may wish to include it here. See this CATLR article on "Developing Rubrics" and, for examples, follow the link in the article to "Rubrics: Advantages and Best Practices."]

### Assignment Descriptions and Grade Breakdown

| Component / Assignment                       | Due Dates                         | Percentage of Grade |
|----------------------------------------------|-----------------------------------|---------------------|
| Pre-assessment quiz *complete for full grade | W1                                | 1%                  |
| Discussion Posts (Online)                    | W1 (2), W2, W3, W4, W11, W14, W15 | 8%                  |
| Discussions in class                         | W1, W2, W3, W8, W12, W7, W11      | 15%                 |
| Participation in class activities            | W1, W4, W8                        | 19%                 |
| At-home Exercises                            | W7, W9, W12                       | 3%                  |
| Project part 1: Presentation and Report      | W5                                | 21%                 |
| Project part 2: Presentation and Report      | W9                                | 16%                 |
| Participation in Tutorial sessions           | W12                               | 2%                  |
| Project part 2: Presentation and Report      | W9                                | 16%                 |
| <b>Total</b>                                 |                                   | <b>100%</b>         |

*Detailed assignment descriptions and rubrics will be provided at least two weeks before due dates. Most of the assignments, including the project instructions and expectations with examples, will be published by the second day of class.*

*This diverse assessment approach hopefully ensures multiple pathways to demonstrate your understanding.*

### Schedule of Meetings and Assignments or of Modules

| Week   | Course focus                                                      | Due assignments                                                                                           |
|--------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Week 1 | Presentation Course/ Pass a problem activity                      | Pre-assessment quiz *complete for full grade; Initial Feedback survey; Discussion Posts (Online/In-class) |
| Week 1 | Modeling Equitable Systems                                        | Discussion Posts (Online); Discussions in class                                                           |
| Week 2 | Modeling Equitable Systems                                        | Discussions in class                                                                                      |
| Week 2 | Modeling Equitable Systems - Case study                           | Discussion Posts (Online)                                                                                 |
| Week 3 | Multiple Objective Decision Making                                |                                                                                                           |
| Week 3 | Multiple Objective Decision Making                                | Discussion Posts (Online)                                                                                 |
| Week 4 | Data Envelopment Analysis - Thinking - Aloud Pair Problem-Solving |                                                                                                           |
| Week 4 | Data Envelopment Analysis                                         | Discussion Posts (Online)                                                                                 |
| Week 5 | Presentations - Project Part I                                    | Peer evaluation (online)                                                                                  |
| Week 5 | Presentations - Project Part I                                    | Peer evaluation (online)                                                                                  |
| Week 6 | Presentation for multiple stakeholders - Project Part I           |                                                                                                           |

|         |                                                                          |                                      |
|---------|--------------------------------------------------------------------------|--------------------------------------|
| Week 6  | Public Health Preparedness and Response Model                            |                                      |
| Week 7  | Public Health Preparedness and Response Model                            |                                      |
| Week 7  | Interdiction and Attacker-Defender - Simulation Activities/ Stakeholders | At-home Exercises/ Jupyter Notebooks |
| Week 8  | Interdiction and Attacker-Defender                                       |                                      |
| Week 8  | Multicriteria and Ranking Models - Game: Spot the limitations            | At-home Exercises/ Jupyter Notebooks |
| Week 9  | Presentations - Project Part II                                          | Peer evaluation (online)             |
| Week 9  | Presentations - Project Part II                                          | Peer evaluation (online), report due |
| Week 10 | Rethink the Multiple Objective Decision Making                           |                                      |
| Week 10 | Rethink the Data Envelopment Analysis                                    |                                      |
| Week 11 | Rethink the Public Health Preparedness and Response Model                | Simulation Activities/ Feasibility   |
| Week 11 | Rethink the Interdiction and Attacker-Defender                           |                                      |
| Week 12 | Rethink the Multicriteria and Ranking Models                             | At-home Exercises/ Jupyter Notebooks |
| Week 12 | Tutorial sessions - in person                                            | Peer (in group) and Instructor       |
| Week 13 | Tutorial sessions - in person                                            | Peer (in group) and Instructor       |
| Week 13 | Tutorial sessions - in person                                            | Peer (in group) and Instructor       |
| Week 14 | Presentations - Project Part III                                         | Peer evaluation (online)             |
| Week 14 | Presentations - Project Part III                                         | Peer evaluation (online), report due |

### Minimum Grade Requirement

A minimum grade requirement: C

Last days to drop the class with a W grade: September?

Last days to drop the class without a W grade: October?

### Late Submissions Policy

Life happens. If you need an extension, please communicate before the deadline. We'll work together to find a solution. Late penalties will apply if not communicated, which is fine. But we cannot accept late assignments after Week 13.

You have the responsibility of choosing what you can and can't do and when.

### Attendance Policy

Your presence enriches our discussions, and our practical activities. Most classes cannot be replaced, since they can only happen in a group. If you miss the class, please contact your peers for notes and see me to discuss key concepts.

Please note the following catalogue policies: *Students have the right to a limited number of excused absences for conditions listed in the Northeastern University Attendance Requirements, including absences due to specific university-sponsored activities, religious holidays, military deployment, and jury duty. Students are responsible for notifying instructors in writing when facing an extended leave of absence or extenuating circumstances. Please note that University Health and Counseling Services will not issue documentation of students' illnesses or injuries.*

### **How We Will Work Together:**

This course runs on conversation. I genuinely want to hear from you throughout the semester, not just on course evaluations but in our everyday interactions. Your feedback helps me adjust things, so they work better for everyone.

I'm hoping we can create the kind of classroom where people actually listen to each other, where we respect the time and energy everyone puts in, and where we can enjoy tackling these complex problems together. Yes, some of this material is challenging, but there's something exciting about working through it as a group.

Here's something I want to be upfront about: I'm beginning my teaching journey with you. This means I'm learning too—not just about teaching, but from all of you. Your questions might catch me off guard sometimes, your perspectives will teach me new ways to see this material, and your feedback will help me become the teacher I want to be. Even years from now, when I'm more experienced, I hope I'll still have this same openness to learning from my students.

So, let's make this a space where we can all be curious, make mistakes, support each other, and figure out together how mathematical modeling can make a difference in communities. We're in this together, and that includes me.

### **How We Will Communicate**

Here is how I plan to use the Canvas communication tools:

**Announcements:** I will post announcements at the end of each week. I will also use the announcements tool to communicate any important or time sensitive updates in the course. Please turn your announcement notifications on in your Canvas setting (Links to an external site.)s so you also receive announcements in email. This will help you keep on top of late-breaking developments.

**If you have questions about content or assignments:** I have set up a discussion called "General Course Q&A". Please post questions about content and assignments here, so your classmates can benefit from the answers. I recommend that you Subscribe (Links to an external site.) to the forum so you also receive forum messages in email.

**Email:** Feel free to email me with questions of a personal nature. You can reach me at [deazevedodrummond.p@northeastern.edu](mailto:deazevedodrummond.p@northeastern.edu) through your NU email account, or you can click **Inbox** in the course menu, search for our course and locate me. See Instructions for how to use the Canvas Inbox here (Links to an external site.). You can expect a response within 24 hours on weekdays, at most 48 hours. Please check your Northeastern email daily or forward it to an email address that you do check.

**"Off and On Office Hours":** I am available for consultations over the phone or through videoconference such as Zoom. Please email me to schedule an appointment outside office hours.

### **Statement on AI Use**

AI tools (like ChatGPT, Claude, etc.) can support your learning when used thoughtfully. And think about how much more rewarding it will be to *connect with your peers before connecting with the AI. They probably can help you with more than you think, and you get the bonus of strengthening your network/friendship circle!*

In this course:

#### **Encouraged uses:**

Getting discussion points mathematical concepts

"Tutoring" your work, asking for clarifying questions or self-assessment tools to help your leaning process.

Improving writing clarity

Debugging code

#### **Not permitted:**

Submitting AI-generated work as your own

Using AI to complete assessments without understanding

Bypassing the learning process

*Remember: The goal is to develop YOUR capabilities. AI is a tool, not a substitute for your thinking. If you are in doubt, come talk to me.*

### **NU Academic Integrity Policy**

Northeastern University is committed to the principles of intellectual honesty and integrity: the

NU Academic Integrity Policy can be found on the website of the Office of Student Conduct and Conflict Resolution (OSCCR).

Just as we value diverse perspectives in our classroom, we also value the authentic contributions each of you makes to our collective learning. Academic integrity forms the foundation of meaningful scholarship and mirrors the ethical responsibilities you'll carry as engineers, especially serving in public programs.

Integrity for this class means celebrating your own unique voice while properly acknowledging when you build on others' ideas; embracing collaboration by listening and sharing our thoughts; asking for help whenever in doubt, and Learning from mistakes and using them as opportunities for growth. In summary, work together, learn from others, and CITE CITE CITE!

Remember that developing your own analytical and problem-solving capabilities is essential for addressing the complex public challenges we study. When you submit work that authentically represents your learning journey, you build the skills needed to serve communities effectively.

If you're struggling with an assignment or tempted to take shortcuts, **please reach out**. Together we can find strategies to support your success while maintaining the integrity that our field—and the communities we serve—deserves.

If you're unsure whether something requires citation, please ask—I'm here to help you learn conventions, and the writing center can also help you.

### **Title IX Protections and Resources**

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal funding. Any NU community member who has experienced such discrimination, sexual assault, relationship violence, stalking, coercion, and/or sexual harassment, is encouraged to seek help. Confidential support and guidance can be found through University Health and Counseling Services, the Northeastern Center for Spirituality, Dialogue, and Service, and the Office of Prevention and Education at Northeastern (OPEN). Note that faculty members are considered “responsible employees” at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator. For additional information and assistance please see the Office of Institutional Diversity and Inclusion webpage.

### **Other resources for your success:**

Make use of all the university has to offer you! And if you need help navigate those services, you can ask for my help.

### **Disability Resource Center**

The university's Disability Resource Center works with students and faculty to provide students who qualify under the Americans With Disabilities Act with accommodations that allow them to participate fully in the activities at the university. Ordinarily, students receiving such accommodations will deliver teacher notification letters at the beginning of the semester. Students have the right to choose whether to disclose their specific disabilities to instructors but must provide a letter to receive accommodations.

### **WeCare**

WeCare offers supports for students during times of difficulty or challenge. You can find WeCare at 226 Curry Student center Monday - Friday from 8:30-5:00, call at 617-373-7591, or email [wecare@northeastern.edu](mailto:wecare@northeastern.edu).

### **Mental Health Resources**

In addition to mental health resources available through Northeastern's University

Health and Counseling Services Northeastern has added Find@Northeastern, which is a 24/7 mental health consulting line and can be reached at 1-877-223-9477.

### **The Writing Center**

The Northeastern University Writing Center offers free and friendly tutoring for any level of writer, including help with conceptualizing writing projects, the writing process, and using sources effectively. Currently, the Writing Center has virtual appointments only. To make an appointment, or learn more about the Writing Center, visit

<https://www.northeastern.edu/writingcenter>, or email [WritingCenter@northeastern.edu](mailto:WritingCenter@northeastern.edu).

Advance and same-day appointments are available.

### **Peer Tutoring**

The Peer Tutoring Program offers a wide range of tutoring services to meet the academic needs of undergraduate students. If you need academic assistance, contact the Peer Tutoring Program Monday through Friday from 9:00am to 5:30pm. Peer tutoring services are free and open to all NU undergraduate students. Peer tutoring begins the second week of classes and ends the last day of classes. The Peer Tutoring Program is located in 1 Meserve Hall. Call 617- 373-8931, email [peertutoring@northeastern.edu](mailto:peertutoring@northeastern.edu), or visit the weblink above.

### **International Tutoring Center**

The International Tutoring Center (ITC) provides current Northeastern University international students with free, comprehensive English language and academic support. The ITC includes English as a Second Language Tutoring (ESL), Language and Culture Workshops, and Reading Workshops. For more information on available workshops and tutoring opportunities please visit the ITC weblink above.

### **Snell Library**

Snell Library offers a variety of resources for undergraduate research, including subject-specific Research Guides, help with citation and bibliography, and 24/7 chat support . The library also houses the Digital Media Commons, which offers a variety of resources for instructors and students for multimedia projects.

### **Final statement:**

This syllabus is our initial agreement, but learning is dynamic, and the classes and syllabus are too. I welcome your feedback throughout the semester on how we can improve the course. Your success is my primary goal, and I'm here to support your journey in mastering the key concepts of Systems Engineering for Public Programs.

*Questions? Concerns? Ideas? Want to chat? My door is always open.*

| Schedule of Meetings and Assignments or of Modules |      |              |                            |                   |                  |
|----------------------------------------------------|------|--------------|----------------------------|-------------------|------------------|
| Week                                               | Date | Course focus | Expected Learning Outcomes | In-class Activity | At-home activity |
|                                                    |      |              |                            |                   |                  |

|        |        |                                                                                                                |                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                               |                                                                                                         |
|--------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Week 1 | 4-Sep  | <b>Presentation Course</b>                                                                                     | Get to know each other. Set expectations between you (the students) and me (the professor). Negotiation and communication skills. Self-assessment skills                                   | Presentation of the class structure and expected learning outcomes.                                                                                                                                                                                                                           | Reading 2 resources + 1 video Pre-assessment quiz *complete for full grade; Initial Feedback survey;    |
| Week 2 | 9-Sep  | <b>Overview of the Models Modeling Equitable Systems</b>                                                       | Identify the characteristics of Modelling Equitable System, Present an overview of Equitable OR framework Present characteristics of Equitable Systems                                     | Mini-lecture on OR framework in social context and stakeholders' importance.                                                                                                                                                                                                                  | Assess a Model by its Equitable considerations Discussion Posts (Online)                                |
| Week 2 | 11-Sep | <b>Modeling Equitable Systems</b> Case study                                                                   | Learn the skill to extract information from multiple sources Understand and critically evaluate different sources content. Hone self-assessment skills                                     | Mini-lecture on Models. Analysis of case study in 3 parts, class; in group and as a class discussion                                                                                                                                                                                          | Mini-lecture on how to read a Case study Project Case study material (part I) Discussion Posts (Online) |
| Week 3 | 16-Sep | <b>Multiple Objective Decision Making</b> Main concepts: Multiple perspectives in decision-making Stakeholders | Stakeholders and their relation to decision-making Consider multiple perspectives in decision-making Design adequate questions to the problem based on stakeholders and expected outcomes. | Exercise in class – Stakeholders Self-explanation of exercise at home (posted)- Pass a problem activity with stakeholders and consequence of decision- Discussion of potential outcomes and impacts on each stakeholder (It will be part of the report, but discussion will be made in group) | Mini-lecture on different (non-obvious) stakeholders in decision Project Case study material (part II)  |

|        |        |                                                                                                                                                                                      |                                                                                                                                                                                                                                                 |                                                                                                                                       |                                                                                                                                          |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Week 3 | 18-Sep | <b>Multiple Objective Decision Making</b> Case study <b>Main concepts:</b> Consequences of decisionsConsequences of different perspectives to modeling decisions in social settings. | Identify at least one unintended consequence of the modeling choices.Identify the trade-offs between different decisions and strategies.Offer at least one alternative modeling strategy.Adapt decision-model strategies to different contexts. | Discussion on Identity (Character building)4S Activity – Decision outcomes in the real world (brainstorm, voting, discussion, voting) | Project Case study material (part II)Post: 200 words post about what is the most important thing when listing assumptions for our models |
| Week 4 | 23-Sep | <b>Data Envelopment Analysis (DEA)</b> <b>Main concepts:</b> DEA Assumptions, Limitations and use of the model in public sectorsData limitationsUnintended Consequences              | Understand the limitations of data beyond the modelUnderstand that actions might have unintended consequencesUnderstand Efficient frontier                                                                                                      | Thinking - Aloud Pair Problem-Solving                                                                                                 | Discussion of expected resultsMock up presentation                                                                                       |
| Week 4 | 25-Sep | <b>Data Envelopment Analysis (DEA) - Case study</b> <b>Main concepts:</b> Data limitationsUnintended Consequences                                                                    | Systematically think how to mitigate effect of limitations and unintended consequences from our modelling effortsUse efficient frontier                                                                                                         | Mock up presentation                                                                                                                  | Define final assumptionsDiscussion Posts (Online) - Peer Feedback on Mock-up presentations and reports draft                             |
| Week 5 | 30-Sep | Presentations - Project Part I                                                                                                                                                       | Give and receive feedbackPresent and defend your choices                                                                                                                                                                                        | Presentation                                                                                                                          | Report;Peer evaluation (online post)                                                                                                     |
| Week 5 | 2-Oct  | Presentations - Project Part I                                                                                                                                                       |                                                                                                                                                                                                                                                 | Presentation                                                                                                                          |                                                                                                                                          |
| Week 6 | 7-Oct  | Presentation for multiple stakeholders - Project Part I                                                                                                                              | Empathy Negotiation skills                                                                                                                                                                                                                      | Presentation of multiple stakeholders                                                                                                 |                                                                                                                                          |

|         |        |                                                            |                                                                            |                                     |                                      |
|---------|--------|------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|--------------------------------------|
| Week 6  | 9-Oct  | Public Health Preparedness and Response Model              | Understand importance of timing in decision-making                         |                                     |                                      |
| Week 7  | 14-Oct | Public Health Preparedness and Response Model (case study) | Understand how power-dynamics can affect decisions and outcomes            | Simulation Activities/ Stakeholders |                                      |
|         | 16-Oct | Interdiction and Attacker-Defender                         | Identify limitations of different modelling strategies using one case      |                                     | At-home Exercises/ Jupyter Notebooks |
| Week 8  | 21-Oct | Interdiction and Attacker-Defender (case study)            | Be capable of choosing among different options using evaluation strategies | Game: Spot the limitations          |                                      |
| Week 8  | 23-Oct | Tutorial sessions - in person                              | Understand assumptions, limitations and use of Interdiction Models         |                                     | Peer (in group) and Instructor       |
| Week 9  | 28-Oct | Tutorial sessions - in person                              | Skill: evaluating and improving work                                       |                                     | Peer (in group) and Instructor       |
| Week 9  | 30-Oct | Presentations - Project Part II                            | Give and receive feedback                                                  | Presentation                        | Peer evaluation (online)             |
| Week 10 | 4-Nov  | Presentations - Project Part II                            | Present and defend your choices                                            | Presentation                        | Peer evaluation (online), report due |
| Week 10 | 6-Nov  | Multiple Objective Decision Making                         | Rethink model                                                              |                                     |                                      |
| Week 11 | 11-Nov | USA: Veterans Day, no classes                              | Application and code                                                       |                                     |                                      |
| Week 11 | 13-Nov | Data Envelopment Analysis                                  | Be able to explain how you are rethinking the model                        | Lecture                             | Mini-Project in Python               |
| Week 12 | 18-Nov | Public Health Preparedness and Response Model              | Apply and code the model with help of support materials                    | Mini-Project in Python              | Simulation Activities/ Feasibility   |

|         |        |                                     |                                                                                                            |                                                                        |                                               |
|---------|--------|-------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------|
| Week 12 | 20-Nov | Interdiction and Attacker-Defender  | Be able to explain how you are rethinking the modelApply and code the model with help of support materials | LectureMini-Proje<br>ct in Python                                      |                                               |
| Week 13 | 25-Nov | Multicriteria and Ranking Models    | Be able to explain how you are rethinking the modelApply and code the model with help of support materials | LectureMini-Proje<br>ct in Python                                      |                                               |
| Week 13 | 27-Nov | Tutorial sessions -<br>in person    | Skill: evaluating<br>and improving<br>work                                                                 | Pair of groups<br>discussion +<br>Instructor<br>orientation<br>session | At-home<br>Exercises/<br>Jupyter<br>Notebooks |
| Week 14 | 2-Dec  | Tutorial sessions -<br>in person    |                                                                                                            |                                                                        |                                               |
| Week 14 | 4-Dec  | Tutorial sessions -<br>in person    |                                                                                                            |                                                                        | Discussion post                               |
| Week 15 | 6-Dec  | Presentations -<br>Project Part III | Give and receive<br>feedbackPresent<br>and defend your<br>choices                                          | Presentation                                                           | Peer evaluation<br>(online)                   |
| Week 16 | 8-Dec  | Presentations -<br>Project Part III |                                                                                                            | Presentation                                                           | Peer evaluation<br>(online), report<br>due    |
|         |        |                                     |                                                                                                            |                                                                        |                                               |